

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, TypeScript, C++, SQL

Tools/Frameworks: TensorFlow, PyTorch, FastAPI, Flask, Docker, Kubernetes, AWS SageMaker, Vertex AI, GitHub Actions, Angular, Nest.js

Competence: Generative AI, Agentic AI, LLMOps, Machine Learning, Natural Language Processing, MLOps, CI/CD, SaaS Development, Data Engineering, Cloud Computing, Performance Optimization, Data Security

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated deep learning models for metrological applications, improving model performance and training efficiency.
- Explored novel architectures and physics-informed hybrid loss functions to enhance model accuracy and stability.
- Collaborated with PhD researchers for a publication at the FLOMEKO conference.
- Built dashboards for data visualization and model performance monitoring.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Improved vitiligo segmentation accuracy to 94% and increased speed by 5x using AI algorithms; deployed with Docker and FastAPI.
- Developed a real-time tennis foul detection system with TensorRT, reducing compute cost and time by 6x.
- Managed MLOps pipelines on AWS SageMaker, addressing YOLO model data drift, and maintained Kubernetes infrastructure.
- Conducted educational sessions on machine learning and big data at universities.
- Developed POCs using YOLO, few-shot learning, and CNN for client demos.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed a full-stack website providing AI-based services using Flask, HTML, and CSS.
- Built an AI chatbot application using OpenAI API and relevant LLM tools/frameworks.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Built web scrapers and ETL pipelines for structured and unstructured data.
- Automated data ingestion and cleaning workflows for client analytics use cases.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a weighted-probability ensemble integrating DPFunc and DeepGOPlus, using AlphaFold for feature extraction.
- Increased model accuracy by 4% through custom feature engineering across MF, CC, and BP namespaces.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed a real-time Bitcoin price forecasting system using LSTM models and Flask for deployment.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Collaborated to build the world's first AI-driven self-playing Chitralli Sitar, combining signal processing and music generation LLMs.
- Project featured on Coke Studio in a music video by Hassan Raheem.

Hugging Face

Open-Source

Contributor

- Co-authored Zero-shot Image Classification documentation and implementation page on the Hugging Face Hub.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Focused on Generative AI, LLMOps, and Natural Language Processing.

National University of Computer and Emerging Sciences

Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor.
- Specialized in Machine Learning and Data Engineering.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python